

ABHINAV TARIGOPPULA

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [LeetCode](#) | [ResearchGate](#)

SUMMARY

Final-year B.Tech CSE (AI & ML) student at GITAM University with three AI/ML internships, including PwC. I build and ship **machine learning, LLM and RAG systems** end to end, from data pipelines and model fine-tuning to **FastAPI** services and React front ends. **29 pull requests merged into 18 open-source organizations** including unsloth, HuggingFace, pydantic, Black, Celery, redis-py and Apache Maven (400k+ combined GitHub stars). Research paper accepted at **IEEE ISED 2026**, NIT Warangal. Open to full-time roles in AI/ML engineering, software development and data science.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C++

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Hugging Face Transformers, Fine-Tuning (PEFT, LoRA, QLoRA), NLP, Computer Vision, OpenCV, YOLOv8, SHAP

LLMs and GenAI: LangChain, OpenAI API, Claude API, RAG, Embeddings, Vector Search (FAISS, Pinecone), AI Agents, Tool/Function Calling, Prompt Engineering, LLM Evaluation, Guardrails, MCP

Backend and Web: FastAPI, REST APIs, asyncio, Pydantic, Node.js, React.js, Next.js, HTML/CSS

Databases: PostgreSQL, MySQL, SQLite, DBMS

Tools and DevOps: Git, GitHub Actions, CI/CD, Docker, Kubernetes, AWS, Linux, pytest

CS Foundations: Data Structures and Algorithms, OOP, Operating Systems, Computer Networks

EXPERIENCE

AI and Data Advisory Intern - PwC Launchpad Advisory Program, PwC India | Remote | Feb 2026 - Jul 2026

- Selected for PwC's flagship GenAI and Data program; ranked among top performers, reaching Level 10 (1,500+ XP) in the cohort.
- Developed LLM prompt workflows and enterprise data architecture designs applied to real client advisory scenarios.

AI/ML Engineering Intern - AVAIntern | On-site, Visakhapatnam | May 2026 - Jun 2026

- Developed and evaluated 5+ ML models on datasets of 10,000+ records, reaching up to 92% prediction accuracy through feature engineering and hyperparameter tuning.
- Automated data preprocessing and analysis workflows with Python, Pandas and Scikit-learn, cutting manual data preparation time by 40%.
- Performed EDA and model evaluation across 15+ features, improving model performance by 18%, and presented findings to stakeholders through interactive visualizations.

Machine Learning and AI Intern - OneStop AI | Remote | May 2025 - Aug 2025

- Improved classification accuracy by about 12% (78% to 90%) across 3 production ML models through hyperparameter tuning and architecture experiments.
- Cut inference latency by about 18% (220 ms to 180 ms) by refactoring preprocessing pipelines and applying model quantization.
- Built end-to-end data pipelines (preprocessing, feature engineering, batch inference) with CI/CD for repeatable model delivery, and added monitoring with the senior engineers.

PROJECTS

DocGuard-VLM - Fine-Tuned Vision-Language Model for Document Intelligence | PyTorch, Transformers, PEFT/QLoRA, unsloth, Qwen2-VL | [GitHub](#)

- Fine-tuned Qwen2-VL-2B with QLoRA for receipt field extraction and tampering detection; on 120 held-out examples, raised field-extraction validity from 25% to 92% and forgery-detection F1 from 0.21 to 0.60 over the zero-shot baseline.
- Built a 5-perturbation adversarial evaluation harness (JPEG, noise, blur, rotation, downscale) and found and documented a dataset-seeding bug that was confounding results.

AgentDesk - Multi-Agent AI Assistant for Customer Operations | Python, LLM Tool Calling, RAG, FastAPI, SQL | [GitHub](#)

- A planner agent routes each request to knowledge, data and action agents that use validated tool calls with structured JSON outputs; RAG over policy documents with source citations and per-session memory.
- Added prompt-injection detection, PII redaction, per-agent tool permissions and human approval for large refunds, served through a FastAPI REST API and scored by an evaluation harness for accuracy, latency and cost.

PathForge - AI Placement Platform | Python, FastAPI, LangChain, OpenAI API, FAISS, React | [Live Demo](#)

- Designed and shipped a full-stack platform that generates role-specific study material and mock interviews; owned it end to end from backend REST APIs to the React UI.
- Built a RAG pipeline with a skill-gap scoring layer that filters retrieval by the learner's level, improving answer relevance and cutting token usage by about 30%.

CodeHound - Python Static Analyzer | Python, AST, pytest, GitHub Actions | [GitHub](#)

- Built an AST-based static analyzer with 104 checks for async-safety and correctness bugs (blocking calls in async code, fire-and-forget tasks, resource leaks, mutable defaults); bugs it found were fixed and merged in unsloth, mem0, agno, pydantic, Black and Weaviate.

OPEN SOURCE CONTRIBUTIONS

- **29 PRs merged across 18 organizations:** unsloth (77k stars), mem0 (66k), Black (42k), agno (42k), pydantic (29k), marimo (23k), HuggingFace (huggingface_hub, accelerate, peft), pydantic-ai, Prowler, Snipe-IT, redis-py, scapy, Xorbits Inference, Celery (kombu), pymodbus, Weaviate, Apache Maven and Mercari; 400k+ combined stars, all reviewed and merged by maintainers. [View merged PRs](#)
- Fixed event-loop blocking in async code, connection and file-handle leaks, state shared across connections and objects, unclear errors and timestamp precision, each with a regression test; code shipped in huggingface_hub v1.17.0 on PyPI.
- Under review: Celery, asynccpg, aiokafka, valkey-py, PyTorch Geometric, SHAP, paramiko, vLLM, Apache Airflow, LlamaIndex, Microsoft AutoGen and LiteLLM.

EDUCATION

GITAM University, Visakhapatnam - B.Tech, Computer Science and Engineering (AI/ML) | Aug 2023 - May 2027

CGPA: 8.12 | O grade in Machine Learning, A+ in AI and Deep Learning | Coursework: Machine Learning, Deep Learning, Computer Vision, DSA, DBMS, Operating Systems, Computer Networks

FIITJEE, Visakhapatnam - Class XII (MPC) | 2020 - 2022

RESEARCH

- **Accepted, IEEE ISED 2026** (NIT Warangal): "Machine Learning-Based Student Performance Prediction: A Comparative Study Using Ensemble Methods and Explainable AI" (second author).
- **Abstract accepted, ICTIRL-2026** (Springer Nature - Atlantis Press proceedings): "AI-Assisted Bug Detection and the Accountability Gap: A Case Study of an Open-Source Static Analyzer" (first author).
- **Abstract submitted**, Springer book chapter on lightweight deep learning deployment on edge hardware (first and corresponding author). Preprint: HAPS, a hybrid AI proctoring system using CNNs and YOLO (first author).

CERTIFICATIONS

- Anthropic Academy: Building with the Claude API, Model Context Protocol, Agent Skills, Claude Code in Action; Salesforce Agentblazer Champion 2026; IBM SkillsBuild Code Generation with IBM Granite; Cisco Introduction to Modern AI; Udemy Mastering AI Agents.
- AWS Solutions Architecture Job Simulation (Forage); APSCHE-SmartBridge AI and ML Virtual Internship; Infosys SpringBoard Database and SQL; Nasscom FutureSkills Prime; TCS iON Career Edge.

ACHIEVEMENTS

- Top 5 at the AI and Machine Learning Hackathon by Techgyan Technologies at BITS Pilani, Hyderabad (24-hour, in person); participant in Smart India Hackathon 2024 with SoulSync, an AI mental-health chatbot.
- 212 LeetCode problems solved (91 easy, 108 medium, 13 hard) with an 82-day best streak.
- Student Placement Coordinator (SPOC): point of contact between the placement cell, visiting recruiters and student cohorts for campus drives.